

# Simi Dalo Mwadkwon

Senior Data Scientist · Machine Learning Engineer

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## PROFILE

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Data scientist with eleven years in insurance, public health and edtech, the last six building models that decide something: which claim to review, which learner needs help, which home to show first. Shipped a pricing model that cut loss ratio 4.2 points across 900,000 policies, and a learner risk model now used on 40,000 students a term. Teach tree-based modelling and experimentation to working engineers in 30 countries.

## CORE COMPETENCIES

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### Modelling

- Gradient boosting (XGBoost, LightGBM), regularised regression, survival and uplift models.
- Causal inference for pricing and retention: A/B and switchback tests, difference-in-differences.
- Calibration and ranking metrics — AUC, NDCG, expected loss, not accuracy theatre.

### Engineering & Platforms

- Python, SQL and R in production; PySpark and dbt for pipelines over 400 GB a day.
- ML lifecycle on AWS and GCP: feature stores, batch and online scoring, drift monitoring.
- Reproducible analysis with Quarto, Typst and Git; CI that fails on an unreviewed model change.

### Leadership & Communication

- Led four analysts and one ML engineer; ran hiring, review cycles and the modelling roadmap.
- Translated model risk into decisions for regulators, underwriters and ministries of health.
- Designed and taught an 8-week applied ML curriculum for 300+ learners.

## PROFESSIONAL EXPERIENCE

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### Senior Data Scientist – Northwind Insurance Group, Frankfurt

2022 – Present

Hybrid

- Built the pricing model that replaced 14 regional rate tables — loss ratio down 4.2 points across 900,000 policies in the first year.
- Cut claim-review workload 38% with a triage model at a fixed false-positive budget, freeing six adjuster days a week.
- Put the model lifecycle under CI: every retrain ships with a data contract, a backtest and a monitoring dashboard, so a bad release is caught before pricing changes.
- Ran 40+ A/B and switchback tests on renewal and bundling offers; two of them paid for the whole team's platform budget.

### Data Scientist, Health Analytics – Meridian Health Trust, London

2019 – 2022

London, UK

- Modelled appointment no-shows across 62 clinics; targeted reminders recovered 11,000 appointments a year.
- Rebuilt the reporting pipeline in dbt and SQL, taking the monthly board pack from nine days to under one.
- Co-designed the consent and de-identification layer with clinicians before any data moved — no incident, no retrofit.

### Machine Learning Engineer – Kestrel Analytics, Lagos

2017 – 2019

Lagos, Nigeria

- Shipped a credit-scoring model to a lender's loan officers in 11 weeks, from notebook to monitored endpoint.
- Cut batch scoring from 6 hours to 25 minutes by reworking the feature pipeline — same model, better jobs.
- Mentored three junior analysts into modelling roles; two are still there.

## **Instructor, Applied Machine Learning** – Example Institute, Jos, Nigeria

2018 – Present

Part-time, remote

- Teach tree-based modelling, experiment design and model evaluation to working engineers — 300+ students, 30 countries.
- Wrote the case-study set now used in three cohorts: churn, claims triage and ranking.
- Run A/B tests on our own teaching material to see which explanations actually move assessment scores.

## **Analyst, Actuarial & Data** – Sabine Reinsurance, Zurich

2015 – 2017

Zurich, Switzerland

- Automated the quarterly reserving pack in R, removing 80 hours of manual work per cycle.
- Built the first reproducible reporting standard the team used for regulatory submissions.

## **EDUCATION**

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### **MSc Financial Engineering** – Example University

2013 – 2015

Distance learning

Thesis: expected-loss calibration for imbalanced insurance portfolios.

### **BEng Electrical & Electronics Engineering** – Example Institute, Jos, Nigeria

2006 – 2011

Jos, Nigeria

Graduated with honours; final-year project on signal processing for medical telemetry.

## **PUBLICATIONS**

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Adeyemi, A. and Mwadkwon, S. (2021). Calibration under class imbalance in insurance triage. *Journal of Applied Analytics* 14(2), 55–71.

Mwadkwon, S. (2019). Consent-first health analytics: a practical pipeline. *Proceedings of the Example Health Data Conference*, 220–228.

## **LANGUAGES**

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English — native · German — B2, working proficiency · French — reading

## **CERTIFICATIONS & TRAINING**

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AWS Certified Machine Learning — Specialty (2024) · dbt Analytics Engineering (2023)

Certificate in Teaching and Learning in Higher Education, ref. TF/00412 (2020)